

we can, because our brain curiously interlinks all of the information. IIT sets the amount of interconnectedness as a measure of consciousness.

These abstract postulates are described using principles of mathematics. Given a particular brain structure and all the constituent elements in it, one can, at least in principle, calculate how deeply connected the brain is. This integratedness throws up a single number (ϕ)

scope of the current article; it suffices to say that this relatively new theory does an impressive job.

ENTROPY?

In the same vein but using a highly different approach, earlier this year, scientists proposed a theory of consciousness based on entropy. Entropy put simply is a measure of disorder or randomness of a system and the higher the entropy of a given system the more

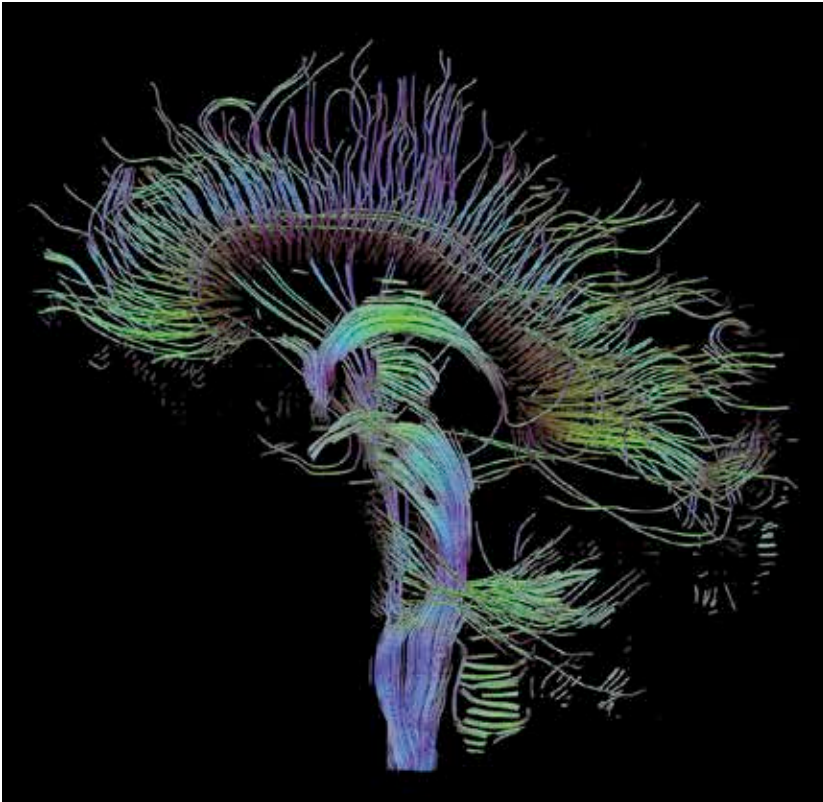
brain following the law of decay and increasing entropy, governed not by the complexity of connections but rather how many different ways these connections between neuron groups can be made – a stark departure from IIT.

Both, IIT and the entropic theory of consciousness use a toolset much different from quantum mechanics to explain consciousness. IIT looks at it as a measure of the connection complexity while the entropic theory makes consciousness as a property that arises from the brain's endeavor to stay within the limits of thermodynamics. Both theories, while less fanciful than the quantum theories, come with their caveats. The study sample size for the entropy study was too small and maybe including other states of mind like drugged and half asleep might help solidify the findings. For IIT, the computation of ϕ , even for simple organisms remains a pipe dream, and the theory predicts that even creatures as simple as a single celled organism will have a non-zero value of ϕ and thus finite consciousness.

COMBINED THEORIES?

There are some who believe that consciousness doesn't care if your favourite topic is physics or math or biology, because all of those are in fact the creations of our consciousness, made up to try and understand the world! It's a bit like a mathematician trying to write a book using only numbers, or a literature major trying to do the same with only quotes from other books... while it's possible to try and do so, a much better book would be one that's written by using whatever language or method is apt at the time. If we are ever to understand consciousness, we're going to have to come up with an explanation that holds true in all fields of study – especially neurobiology and physics.

One thing is for sure, while many might consider the boundaries of uncertainty in their respective fields as the cutting edge of science, or the big questions that remain unanswered, pretty much anyone, scientist or man on the street, will agree that the biggest unanswered question to date is what makes us tick! 



A visualization of the neuron connections in human brain

which tells 'how much' consciousness can a collection of interacting and connecting units subsume. As it is, the theory does tick quite a few boxes. It explains why losing the cerebellum while drastically impairing bodily function yet preserves consciousness, but damage to the cerebral cortex can severely hinder the ability to perceive features of our surroundings we often take for granted – this is because the cerebral cortex is characterized by a more complex brain circuitry and hence a larger value of ϕ . There are some other salient features of consciousness that IIT can give convincing explanations for and though out of the

stable it is. Naturally, systems tend to move towards a state of high entropy, and living systems are in constant decay. The question the theory aims to answer is, "Can we explain consciousness as a consequence of the brain wanting to attain a state of higher entropy?" The scientists looked at how the network of neurons in the brain fired in three stages of an epileptic patient – during wakefulness, sleep and seizure. Results show that the wakeful state is marked by the neurons firing in the largest number of different patterns – indicating a greater number of possible network configurations and thus a higher entropy. The conclusion – consciousness is a byproduct of the